

Chapter 5

Loads on Aircraft

5.1 Question Bank UNIT - 1 AS

Quiz Questions

1. IN a V-N diagram V_s indicates the velocity at which the aircraft can y with load factor
2. Wing loads consist of three types of loads which are: , and
3. 60 degree banked turn requires a load factor of
4. Increased downward load on horizontal tail increases causing the acceleration normal to flight path.
5. Rough runways will cause the load factor to
6. Design gust intensity reduces as the velocity
7. Tail loads contribute to the of the aft portion of the fuselage.
8. When flaps are deployed the maximum design load
9. Load factor be written as
10. Load factor defines load in terms of
11. Aircraft stalling speed increases in proportion to the of the load factor
12. A sudden forward stick movement can cause the load factor to
13. As a design requirement taxiing must be accomplished at
14. The magnification of load factor at the wing tip is due to
15. During banking an increase in bank angle results in an in load factor
16. Lower wing skins are prone to loading than upper wing skins.
17. Wing ribs are planar structure capable of carrying loads
18. Horizontal gusts cause lateral loads on the
19. It is important to calculate gust load factors for an aircraft under weight conditions.
20. Aircraft designed for high altitude flying must be able to withstand loads.

Main Questions

1. Explain Load Factor.
2. Explain with neat sketch wing load distribution due to lift on a wing.
3. Explain the effect of fuselage and wing stores on wing loading.
4. Explain the method of stress analysis for a conventional wing with a neat diagram.
5. Write a short on Corner Velocity in the V-N diagram with the relative equation.
6. Define Limit load, proof Load and Ultimate Load.
7. Draw a typical V-n Diagram and explain salient points.
8. Discuss the general types of loads on aircraft structural components Or What are the major structural components used in aerospace structure?
9. With a neat sketches explain the three different types of loading conditions on Horizontal tail with elevator.
10. Explain the effect of landing and taxi on the wing loads.
11. Draw and explain the V-n diagram for gust envelope mentioning the equation for the gust envelope.
12. Explain the effect of extension of flaps and use of control surfaces on wing loading.
13. With a neat sketches explain the three different types of loading conditions on Vertical tail with Rudder.
14. Draw and explain the V-n Diagram for different Manoeuvres.
15. Explain the loads that act on the landing gears.
16. Explain with a neat sketch structural components wing section and salient features of its structural components.
17. Explain with a neat sketch structural components Fuselage section and salient features of its structural components.
18. An aircraft weighing 6350kgs flying horizontally at velocity of 225m/s was pulled up in a curved path of radius 765m. Calculate the balancing tail load, wing lift and airplane load factor.
19. While flying at 130km/hr (EAS) at a 3km altitude, an aircraft weighing 9kN makes a banked turn through 90 degrees in 15s, maintaining the same incidence and altitude throughout. The wing loading is 100 N/m^2 and the lift to drag ratio is 10 at this speed. Calculate the normal load factor, bank angle, radius of turn and power required.